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Linear Algebra Made Easy and Visual with Python

Linear Algebra Made Easy and Visual | Open-source code & drafts: https://github.com/Visualize-ML

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Why Write "Math Made Easy" Alongside the "Machine Learning" Series?

① All 7 books of the "Machine Learning" series are published, but many beginners found its math "too hard"

② "Math Made Easy" focuses on beginners, breaking down basic concepts to make them fully learnable

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"Math Made Easy" Is a Companion for Learners at Every Stage

① Helps high schoolers build a math foundation for informatics competitions

② "Math + code + visualization" helps undergraduates appreciate the beauty of math while applying it

③ Helps graduates revisit and connect concepts, building a sturdier mathematical mindset

④ Offers a clear path for graduate-entrance exam prep, easing the learning burden

⑤ Helps programmers and coders understand the math behind their code

⑥ A great starting point for beginners in math and coding, including readers from non-STEM backgrounds

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Large Language Models Are Becoming Trusted Learning Companions

① Models like DeepSeek and ChatGPT can organize knowledge, build personalized learning paths, answer questions in real time, and visualize complex ideas

② "Math Made Easy" weaves AI companionship into its teaching ecosystem, encouraging readers to make it part of how they learn

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How "Math Made Easy" Differs from the "Machine Learning" Series

① The "Machine Learning" series ends at machine learning; "Math Made Easy" starts and stays focused on math itself

② The "Machine Learning" series is a sweeping 7-book system; "Math Made Easy" is light and focused, one math domain at a time

③ "Math Made Easy" is simpler and more basic, better suited to beginners, yet covers a more complete knowledge network

④ Here, "code + visualization" is only a means of unpacking math concepts, never an end in itself

⑤ Even for shared topics, the content is entirely rewritten and most figures are recreated from scratch

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In the AI Era, Math Isn't Obsolete — It Matters More Than Ever

① Math underlies how we understand the world, and it's the logic driving large language models like DeepSeek and ChatGPT

② As models grow more capable, we need enough mathematical literacy to grasp their principles and spot their limits

③ Understanding beats rote memorization — it lets us engage AI tools as equals, turning math into everyone's core competitive edge

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Coding Is Not the Goal — It's a Tool, a Means to an End

① Coding is becoming the "second language" of the data era, much as English became the passport of globalization

② The goal of learning to code here isn't to become a professional developer, but to turn theory into working solutions more efficiently

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The Solution: Code + Math. The Best Solution: Code + Math + Visualization

① Learning math while verifying it in Python is an efficient way to close the gap between theory and practice

② Visualization turns abstract formulas into vivid geometric pictures in the mind — the signature tool of "Math Made Easy"

③ The coding here emphasizes computation, not drawing; readers wanting advanced visualization coding can turn to the "Machine Learning" series

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Thanks to the Readers and Collaborators Who Made This Possible

① Deep thanks to our readers — your comments, feedback, and criticism are the greatest motivation to keep refining this work

② Special thanks to Mr. Luan Dacheng of Tsinghua University Press, whose steady support made both the "Machine Learning" and "Math Made Easy" series possible

③ A new collaborator joined the team, Cui Ying (Cuicui), who has led production of every companion video for "Math Made Easy"

④ To my parents.

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What Makes "Math Made Easy" Distinctive?

① Direct line to the author across every channel: if something is unclear, that's on the author — please tell me where you got stuck

② No assumed prerequisites, no filtering readers out — it assumes you start from zero

③ A scaffolded, discovery-driven method that connects knowledge into a network, revisiting the same idea from different angles

④ Depth over breadth — thorough explanation over cramming in topics, drilling, or exam tricks

⑤ Visually delightful, applying the image quality of Visual Beauty of Math to reveal the geometry behind simple concepts

⑥ Math dressed to be approachable — as few formulas as possible, explained conversationally

⑦ Every section includes exercises, with guided self-study using DeepSeek, ChatGPT, and other language models

⑧ Completely free and open source — the PDF manuscript, Python code, and videos are all freely available

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Who Is "Math Made Easy" For?

① Readers who find abstract symbols dull but respond to geometric pictures

② Readers who think math can be "fun" and "beautiful"

③ Readers wanting an "aha" moment — more curiosity, more spatial intuition

④ Readers who want more than drilling problem sets and exam tricks

⑤ Readers who gave up on math, or were told to, and want to pick it back up

⑥ Readers who want to combine math, coding, and visualization

⑦ Readers curious what math is actually good for, and how to use it

⑧ Readers learning to code without a clear direction — and strong math readers who enjoy the aesthetics

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Companion Resources

Print Edition
Reviewed and proofread by the publisher; details may differ slightly from the PDF

PDF Manuscript
Convenient for reading on mobile devices, updated alongside the book

Python Code Files
Jupyter Notebooks, available at https://github.com/Visualize-ML

Companion Videos
Fun, substantive videos that build spatial imagination and geometric intuition, on Bilibili — DrGinger

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JupyterLab or Jupyter Notebook Is the Recommended Companion Tool

① Jupyter combines browser, code, documentation, plotting, media, and publishing in one place — ideal for exploratory learning

② It supports running code in blocks, inline results, markdown notes, embedded images and video, and LaTeX formulas

③ Download Anaconda to get JupyterLab, Spyder, PyCharm, and other common tools together: https://www.anaconda.com/

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Feedback and Suggestions Are Always Welcome

① Series contact email: jiang.visualize.ml@gmail.com

② Readers who offer especially valuable feedback receive a small Iris Flower Books gift as a token of thanks

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Linear Algebra Is the Mathematical Language of Modern Science and Engineering

① It bridges single-variable to multivariable calculus, and univariate to multivariate statistics

② It appears throughout machine learning, computer vision, NLP, robotics, and quantitative finance

③ Every graduate should be able to wield linear algebra as a comfortable, everyday tool in study, research, and work

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What's Wrong with How Linear Algebra Is Often Taught Today?

① Geometric intuition is neglected: few 2D/3D visuals for abstract concepts, leaving students unable to picture them

② Computation over understanding: exercises favor rote technique over grasping geometric meaning and modeling

③ Weak ties to real applications: little connection to data analysis, image processing, or machine learning

④ Disconnected from modern tools: pages of formulas with no Python coding or visualization to deepen understanding

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How Linear Algebra Made Easy and Visual Addresses These Problems

① Systematically builds geometric intuition with visuals and animations for vectors, matrices, determinants, eigendecomposition, and more

② Shifts the focus from solving problems to understanding and applying them, via real cases like PCA, image compression, and least squares

③ Structures knowledge sensibly: core tools like matrix multiplication, determinants, inverses, and vector spaces first, advanced topics later

④ Weaves together math, coding, and visualization, using Python, Jupyter Notebook, and language models as teaching tools

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Volume 1 Has 7 Chapters, Covering the Foundations of Linear Algebra

① Vectors: introduced algebraically through an RGB-color example, covering addition, scalar multiplication, normalization, dot product, projection, and norms

② Matrices: common properties, shapes, and basic operations, with a dedicated section on the easily-mistaken matrix transpose

③ Matrix Multiplication: understood through the dot-product, outer-product, and column/row linear-combination viewpoints

④ Determinants: builds geometric intuition starting from how a 2×2 determinant relates to parallelogram area

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From Inverse Matrices to Systems of Equations, a Geometric View Throughout

⑤ Inverse Matrices: understood as the inverse of a geometric transform, built through scaling, rotation, and shearing

⑥ Vector Spaces: continues the RGB color-space example, using parallelogram grids to explain basis and related concepts

⑦ Systems of Linear Equations: treated as an application of matrix multiplication, determinants, and inverse matrices, closing out the volume

⑧ With that, let's begin the journey through Linear Algebra Made Easy and Visual — and make linear algebra truly easy!

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Volume 1 Stocked the Linear Algebra Toolbox — Volume 2 Puts It to Work

① Geometric Transformations: revisits nearly every Volume 1 tool — scaling, rotation, shearing, projection, reflection, translation, affine maps — feeding into later matrix decompositions

② Projection: builds on orthogonal projection through the geometry of orthogonal matrices, Gram-Schmidt, and QR decomposition

③ Regression: explains simple, multiple, and polynomial regression from the projection viewpoint

④ Eigendecomposition: finds a matrix's eigenvectors and eigenvalues, with emphasis on the spectral decomposition of symmetric matrices

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From Dimensionality Reduction to Graphs and Networks, Toward Data Science and ML

⑤ Dimensionality Reduction: uses eigendecomposition on covariance and correlation matrices for reduction, compression, and feature extraction

⑥ Quadratic Forms: starts from positive definiteness, covering Cholesky and LDL decomposition, the Rayleigh quotient, and Mahalanobis distance

⑦ Singular Value Decomposition: compares the full, economy, compact, and truncated forms of SVD and their geometric meaning

⑧ Graphs and Networks: uses adjacency, transition, and Laplacian matrices to explain spectral clustering — and now, let's make linear algebra not just easy, but useful!

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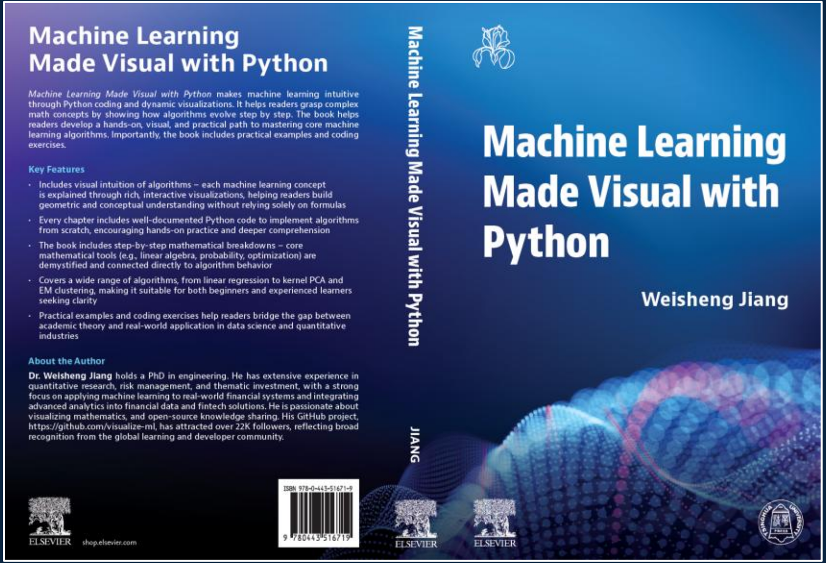
数学

数学基础　线性代数　概率统计

Python编程　数据可视化

工具



数据分析
图和网络

回归、降维
分类、聚类

实践

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